

R E S U M E



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Summary

Senior MES & IoT Engineer (M.Eng. Mechatronics & Cyber-Physical Systems) with over 5 years of experience in the digital transformation of process manufacturing. Expert in IT/OT convergence and ISA-95 compliant MES architecture. Successfully rolled out energy and production management systems across 17 sites and connected over 100 machines via OPC UA and MQTT. Focused on the end-to-end integration of complex manufacturing landscapes, system architecture, and shop floor process automation. DAAD Scholarship recipient.

Work Experience

05.2022 – Current

Deloitte Consulting Solutions GmbH, Leipzig

Senior Solution Engineer

Energy & Production Data Management Rollout 17 Plants

- Led a 2-year global rollout of an energy and production data platform across 17 international plants, establishing real-time transparency for the entire manufacturing network.
- Connected 100+ assets and energy meters via OPC UA, Modbus, and MQTT, replacing manual data entry with automated real-time pipelines.
- Standardized KPI frameworks (OEE, availability, performance, quality) to enable live, data-driven decision-making for executive management.
- Conducted pre-commissioning OT network assessments and hardening; commissioned Siemens S7, Kepware, and Allen-Bradley edge infrastructure and trained local plant engineers on operations and troubleshooting.
- Owned end-to-end programme delivery as central technical stakeholder interface from requirements workshops through configuration to global go-live.

MES Interface Analysis & MES–ERP Integration Strategy

- Reverse-engineered undocumented, in-house MES systems by leading cross-functional workshops with IT and Engineering to create comprehensive interface maps from scratch.
- Assessed all legacy MES–SAP ERP integration touchpoints, identifying compatibility gaps and validating data exchange viability.
- Authored a technical roadmap and API specification, defining REST-based architectures to ensure scalable, maintainable MES–ERP data exchange.
- Supported API development and end-to-end testing, ensuring stability of newly deployed MES–ERP interfaces.

Smart Manufacturing Showcase Factory

- Configured SAP Digital Manufacturing end-to-end: modelled operations, SFC lifecycles, and order execution logic for a fully digital production environment.
- Supported Integration of SAP DM with S/4HANA and EWM, synchronizing order processing and material staging across the full ISA-95 Levels 3–4 stack
- Developed custom SAP DM functions and POD plugins to demonstrate complex, client-specific manufacturing scenarios.

04.2021 – 10.2021

GBneuhaus GmbH, Neuhaus am Rennweg, Germany

Master's Thesis Engineer

- Engineered and simulated an automated cleanroom Pyrosil pre-treatment process using ABB RobotStudio, integrating robot path planning and environmental parameter validation to optimize cyber-physical system performance within a smart manufacturing cell.

- 10.2019 – 03.2022 **International Office, Deggendorf University of Applied Sciences, Deggendorf, Germany**
International Tutor
- Mentored 200+ international students; organised university-level cultural and academic events building cross-cultural communication and stakeholder coordination skills applied directly in multinational client engagements
- 12.2015 – 12.2016 **Thermax Ltd., Pune, India**
Bachelor's student in the field of hybrid coolers
- Enhanced manufacturing efficiency and safety by standardizing hybrid cooler pneumatic testing, implementing a "seamless welding" process to eliminate manual grinding, and authoring a safety manual that reduced shop-floor accidents.

Academics

- 10.2018 – 03.2022 **Deggendorf University of Applied Sciences (Technology Campus Cham), Cham, Germany**
Mechatronic and Cyber-Physical Systems (M. Eng.)
Master's thesis title: Automation of the Pyrosil pre-treatment process for functional spray coating under cleanroom conditions — robot path planning, process simulation (ABB RobotStudio), integration into a controlled smart manufacturing cell.
Grade: 2.0
- 07.2013 – 06.2017 **Savitribai Phule Pune University, Pune, India**
Mechanical Engineering (B. Eng.)
Title of the bachelor thesis: Process Designing for Pneumatic testing of Hybrid Chiller.
Final grade: 1st Class

Other Skills

Certifications

- Professional Scrum Master - I
- AWS Cloud Practitioner
- PTC Kepware Associate Certification
- German Language B2 Certification from Goethe Institute

Technical Skills

- **MES & MOM:** MES Architecture (ISA-95/88), SAP Digital Manufacturing, OEE/KPI Frameworks, MES–ERP Integration.
- **IIoT & Connectivity:** OPC UA, MQTT, REST APIs, PTC Kepware, Siemens S7, Allen Bradley, IT/OT Convergence, IEC 62443, SCADA
- **Cloud & Data:** AWS, Azure, Power BI, Python, JavaScript, Predictive Maintenance
- **Project & Tools:** SAP Cloud ALM, Jira, Agile, Waterfall, Multi-site Rollout Management, Workshop Facilitation, Testing and Test Management
- **Automation & Engineering:** PLC Programming, ABB RobotStudio, AutoCAD, SolidWorks

Language skills

- Marathi (Mother tongue)
- Hindi (local language)
- English (fluent)
- German (fluent / B2)

Scholarship

DAAD Scholarship: Awarded by the German Academic Exchange Service for outstanding academic performance and intercultural engagement during master's studies in Germany.